Public Ledgers

The open nature of public ledgers comes with its own benefits and drawbacks. One major benefit is they are usually open-source. So anyone can take the code of an existing product and make a new version based on it. These new versions are called forks of the original version. The decentralized structure reduces the cost of operations by removing the middleman. This particular fact has become a major disruption in the financing industry. The point of making a currency was to centralize money and now this new technology has questioned the existence of such central bodies. Similar dilemmas are being faced by some other industries that need security of encrypted transactions and large-scale decision making. These kinds of systems enable dodging the bureaucracies. But what takes the titular attention is the fact that there's no central body managing it. This brings the question of security of the transactions. The answer to these questions are the various proofing methods like PoW and PoS. But because both the methods have major drawbacks, concern still looms over security.

Some widely known examples:

Bitcoin - This is an obvious one. The concept of blockchains was introduced to the world by this insanely popular cryptocurrency. This is also the reason why most people are unaware of other important uses of blockchains. Because of Bitcoin, cryptocurrencies have now become the only mainstream use of blockchains.

Ethereum - It is one of the forks of bitcoins and has its very own blockchain system. Albeit based on the one from Bitcoin, but functionally different. Two key differences are its use of smart contracts to be outside the blockchain and modifiable for specific requirements and a recent switch from PoW to PoS protocol.

Private Ledgers

At the organisational level, some operations might require a high level of security and zero room for compromise. Some organisations might even have regulatory compliances that demand security of information and restrict transparent access only to an elected group of people. For such scenarios public ledgers become harmful. But the encrypted nature and faster decision-making abilities of blockchains can still be used by creating a central group. This group controls the blockchain and decides the rights provided to its members. The consensus in such a system completely